6.7 Polar Coordinates

Name _____ Period ____ Date _____

Pre-Calculus 6.7 (30-minute plan) · Student sheet

I can…

- plot points and give several polar representations of a point.
- convert points between polar and rectangular form.
- convert equations between polar and rectangular form.

Vocabulary · fill in the missing word

Pole — the origin of a polar _____ system

Polar axis — the ray from the pole along the _____ x-axis

Polar coordinates — (r, θ): r is the _____ distance from the pole and θ is the directed angle from the polar axis

Part 1 · Notes and examples (fill in each note, then work the example)

Polar coordinates (r, θ): r is the directed distance from the _____; θ is the angle from the polar axis.

Example 1

Plot $(2, \frac{\pi}{3})$, $(3, -\frac{\pi}{6})$, and $(-1, \frac{3\pi}{4})$.

Multiple representations: $(r, \theta) = (r, \theta + 2n\pi) = (_____, \theta + (2n + 1)\pi)$

Conversions: $x = r\cos\theta$, $y = r\sin\theta$; $r^2 = x^2 + y^2$, $\tan\theta =$ _____

Example 2

Convert $(\sqrt{3}, \frac{\pi}{6})$ to rectangular coordinates.

Example 3

Convert $(-1, 1)$ to polar coordinates.

Part 2 · Partner practice

Partner practice

Convert $(0, -5)$ to polar coordinates.